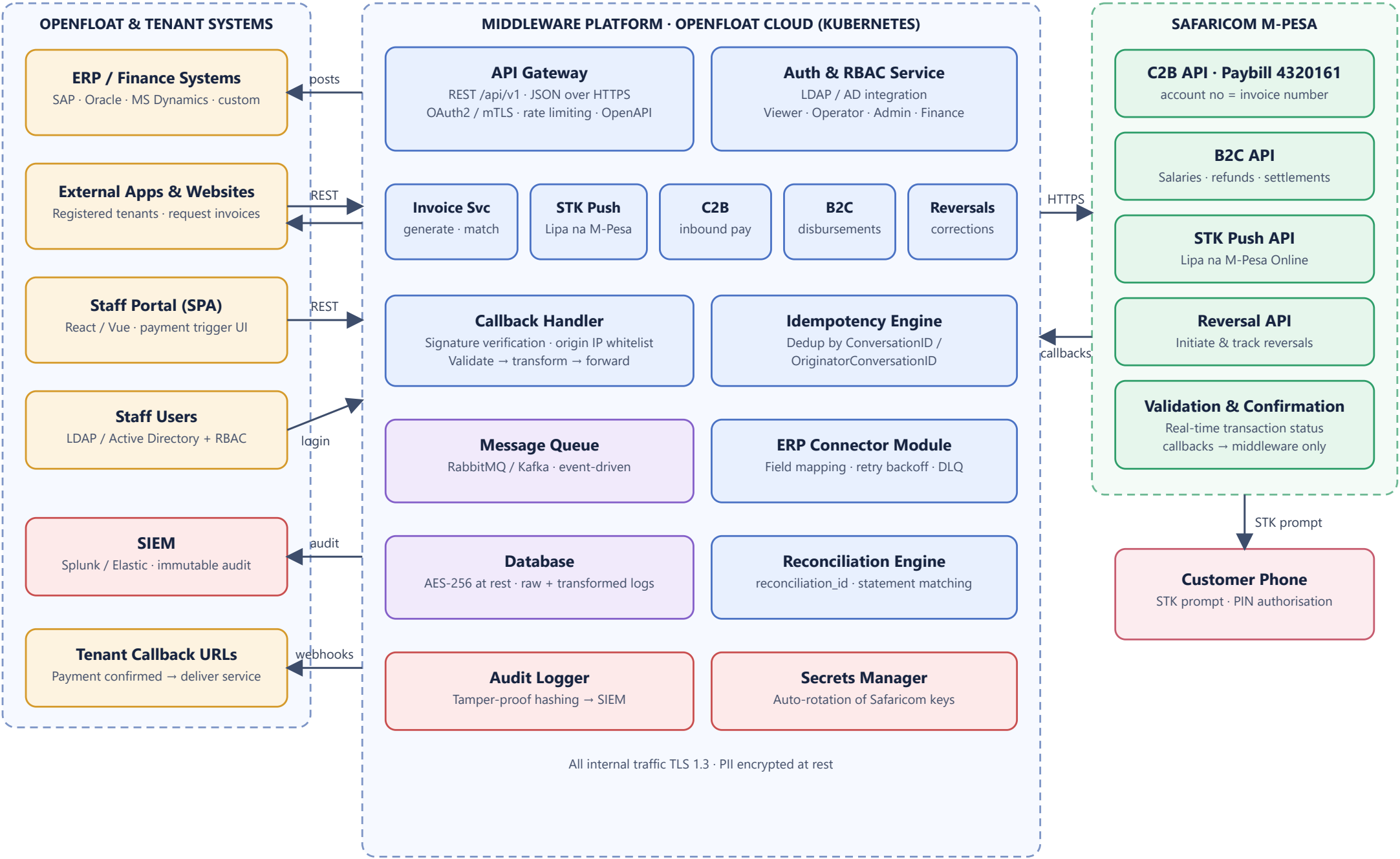


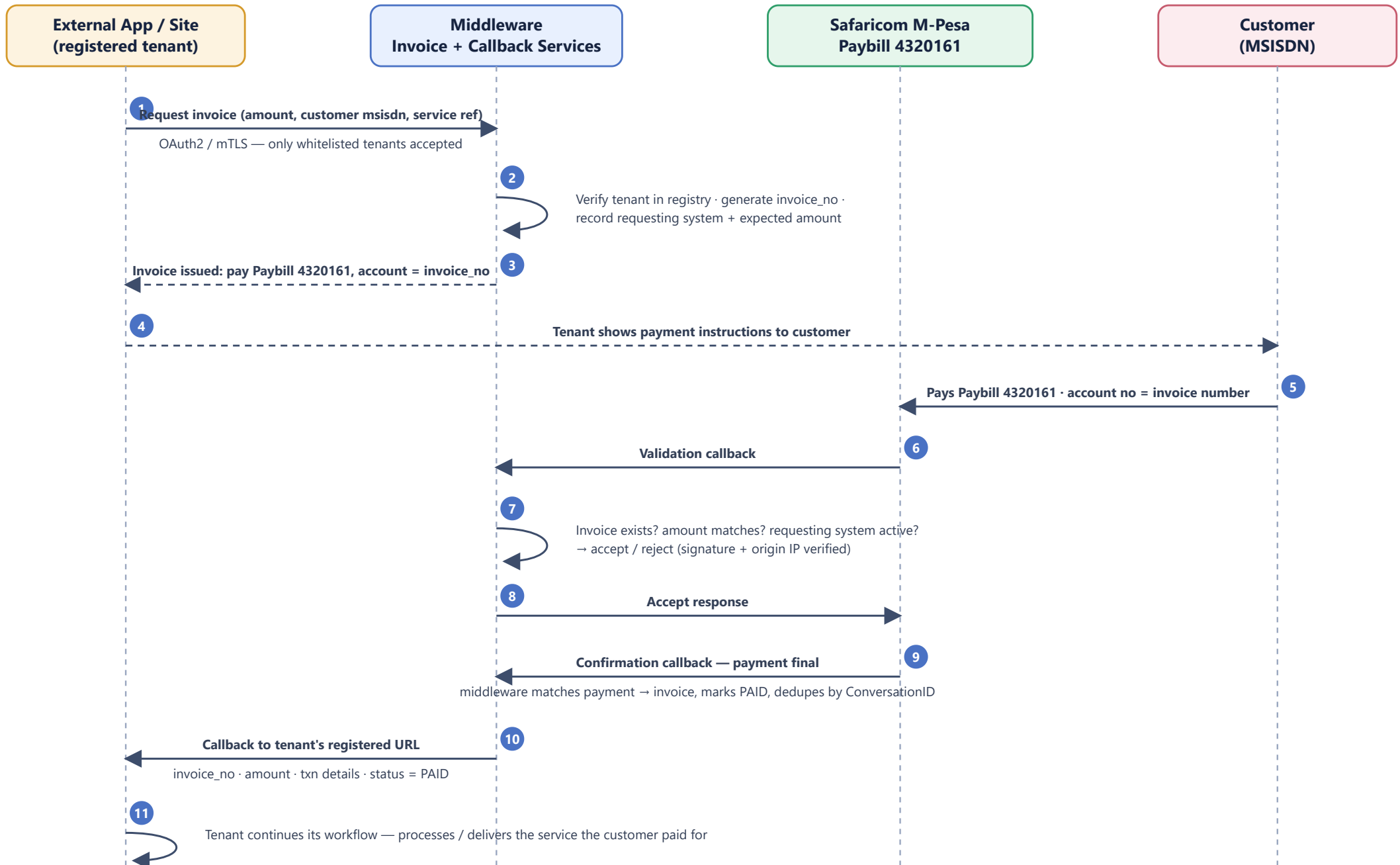
1 · System Component Architecture

OpenFloat M-Pesa Middleware Platform — all M-Pesa traffic flows through a single secure entry point



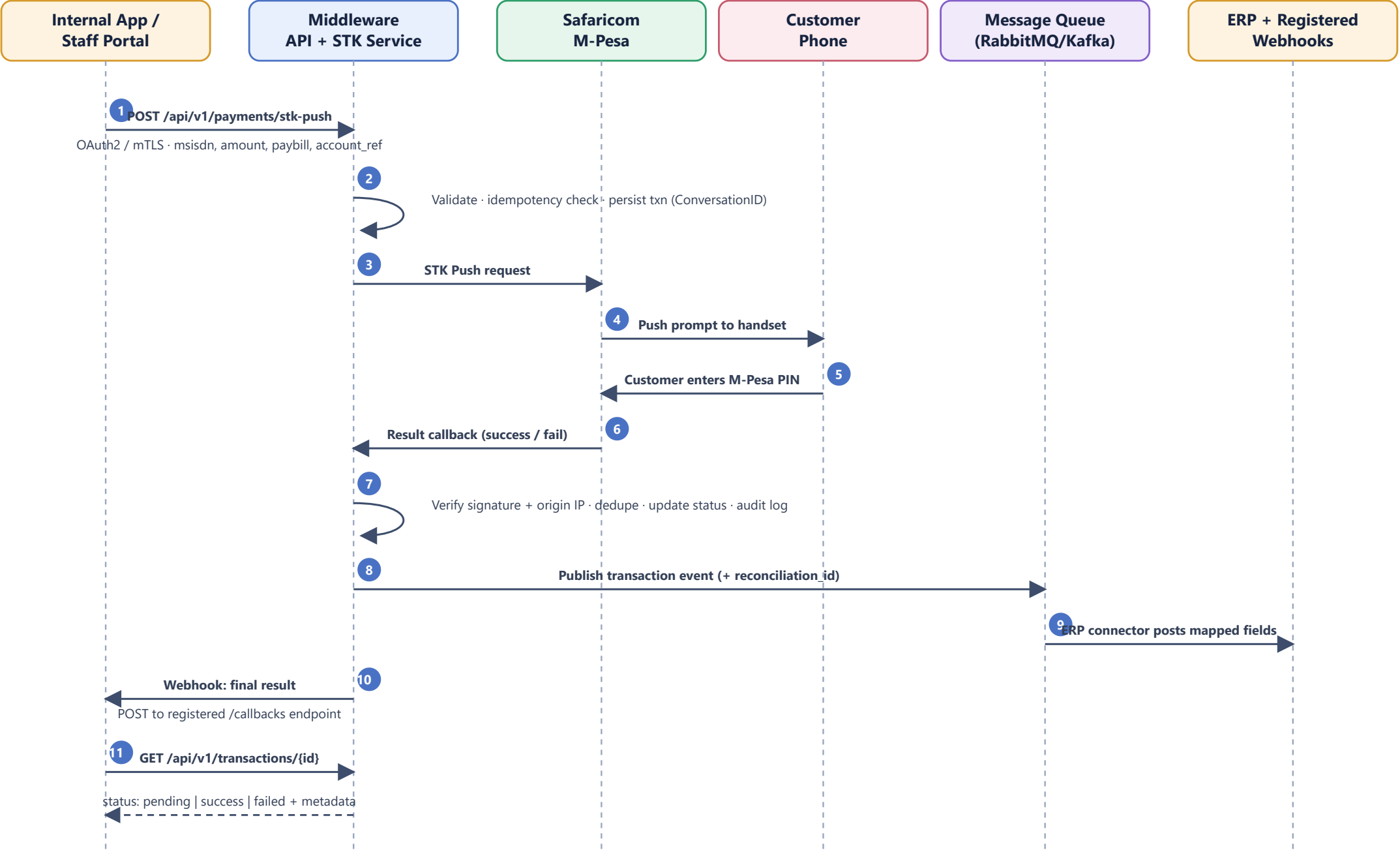
2 · Invoice Payment Lifecycle (Multi-Tenant)

External apps & sites request invoices · customer pays Paybill 4320161 with account = invoice number · confirmed payment triggers a callback so the tenant delivers the service



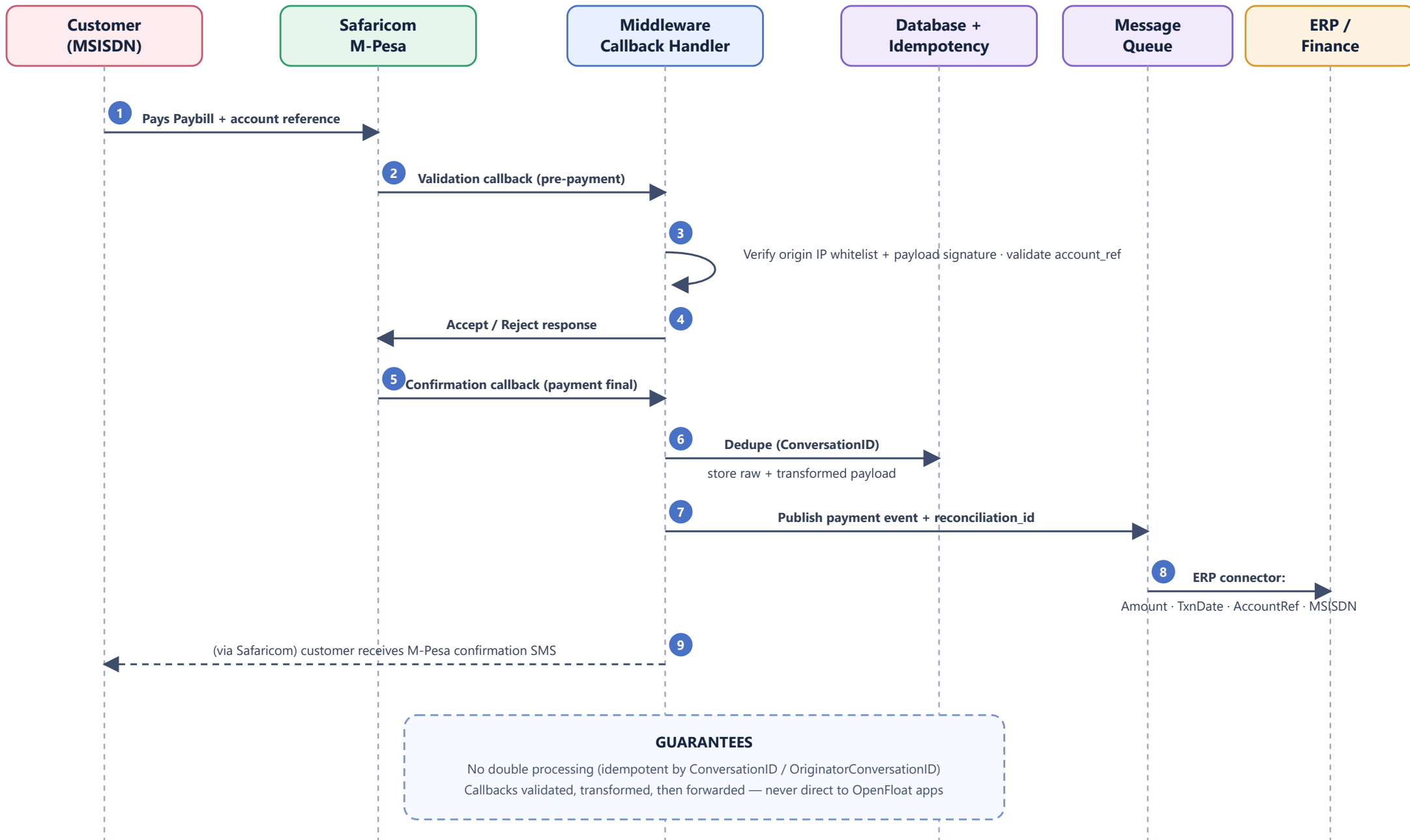
3 · STK Push Payment Flow (Lipa na M-Pesa Online)

End-to-end sequence: initiation → customer authorisation → callback → ERP posting



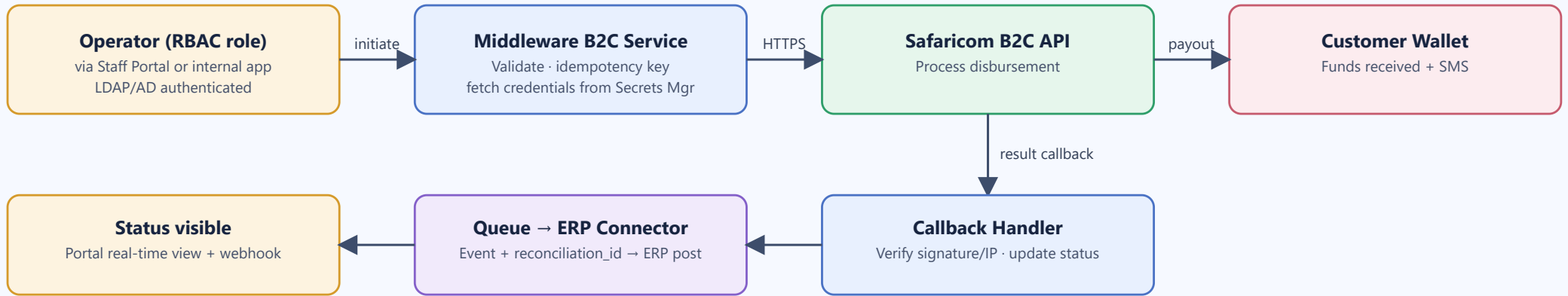
4 · C2B Payment — Validation & Confirmation Callback Flow

Customer-initiated Paybill payment: all Safaricom callbacks land on the middleware, never directly on OpenFloat apps



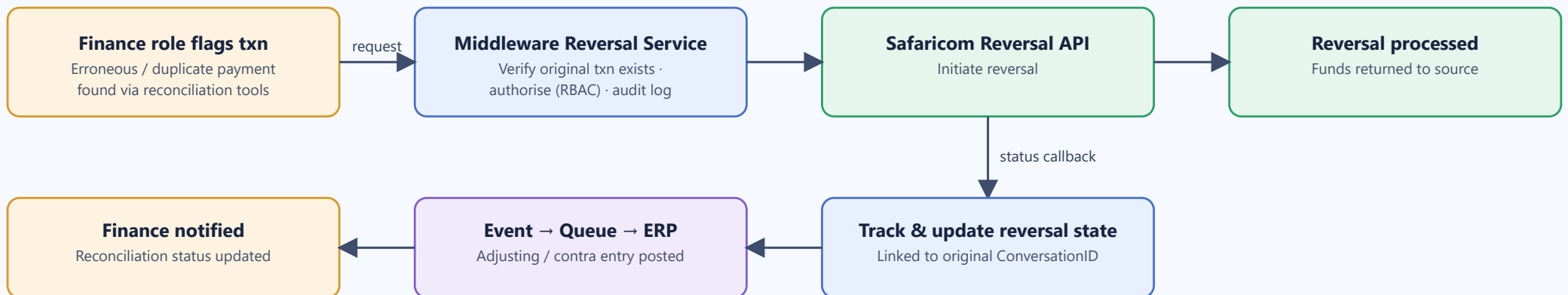
5 · B2C Disbursement & Transaction Reversal Flows

B2C — SALARIES · REFUNDS · SETTLEMENTS



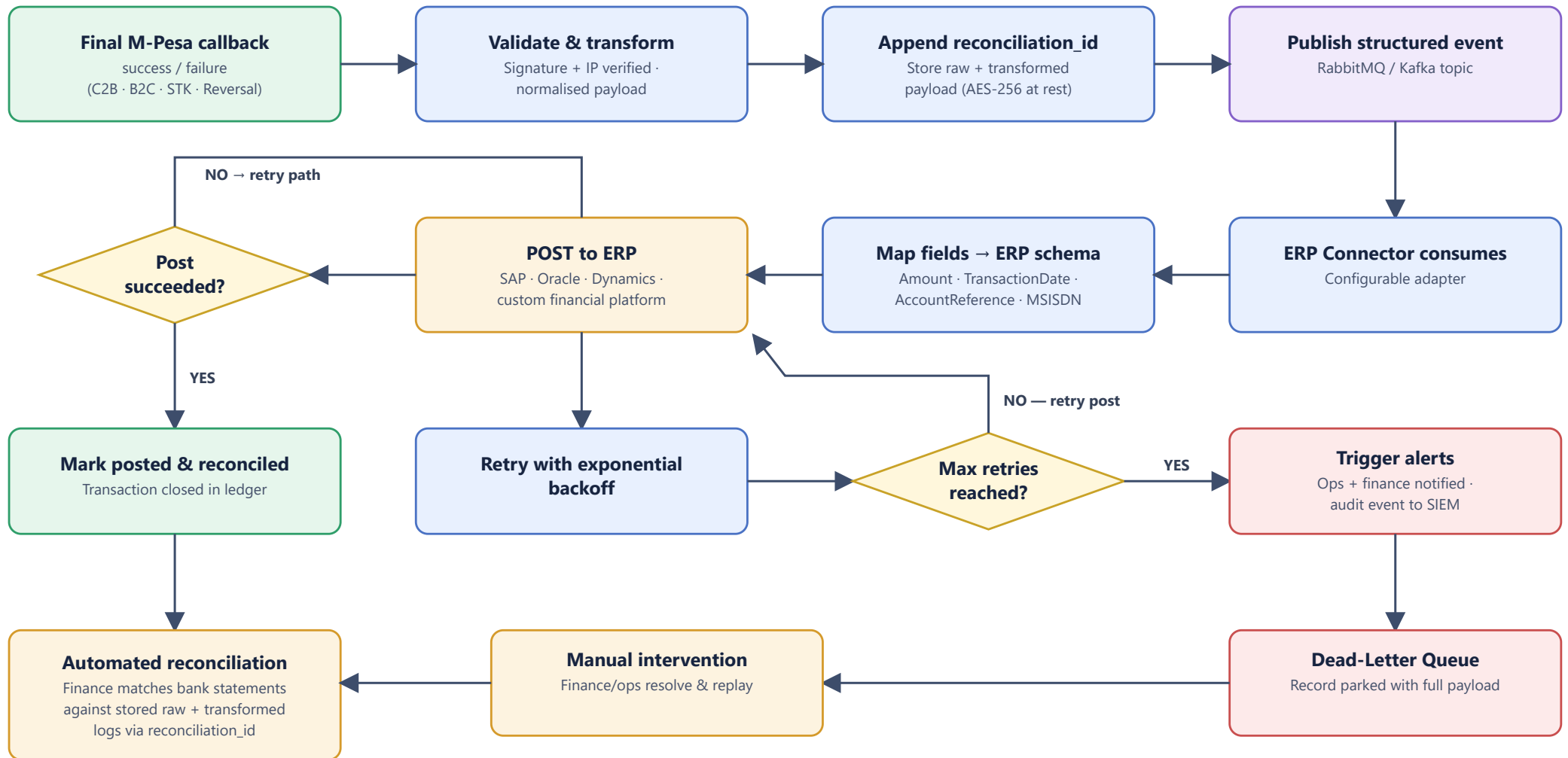
Every B2C request carries a unique OriginatorConversationID — retried submissions are deduplicated, preventing double payouts.

REVERSALS — CORRECTING ERRONEOUS TRANSACTIONS



Reversal lifecycle fully audited: initiator, approval role, timestamps, and final Safaricom status stored immutably.

6 · Real-Time ERP Posting — Event Flow with Retry & Dead-Letter Handling



DELIVERY GUARANTEES

At-least-once event delivery with idempotent ERP posting · every event carries reconciliation_id · no transaction is silently lost — it is either posted, retried, or parked in the DLQ with an alert

7 · Enterprise Authentication & Security Architecture

